

Project Phase 5 — Loan Approval Prediction

Logistic Regression Model — Training & Prediction Report

1. Algorithm Selection

This is a **classification problem**: the target variable, `Loan_Status`, is categorical (Approved = 1 / Rejected = 0), not a continuous number. Per the task rule (Linear Regression for regression problems, Logistic Regression for classification problems), **Logistic Regression** was selected. It directly models the probability of loan approval using a sigmoid function and is well suited to a binary outcome with a mix of numeric and encoded categorical features.

2. Train/Test Split

Split	Rows	Columns (features)	Approval Rate
Training set	491	11	68.6%
Testing set	123	11	69.1%

The cleaned dataset (614 rows x 12 columns) was split 80% train / 20% test using **`train_test_split(test_size=0.2, random_state=42, stratify=Loan_Status)`**. Stratification kept the approval ratio nearly identical between train (68.6%) and test (69.1%) sets, so the model is evaluated fairly. Features were standardized with `StandardScaler` (fit on train, applied to test).

3. Model Training

Model: **`sklearn.linear_model.LogisticRegression(max_iter=1000, random_state=42)`**, trained on the scaled training features. The solver converged in **8 iterations**, confirming the model trained successfully with no errors.

Property	Value
Algorithm	Logistic Regression
Training samples	491
Features used	11
Solver iterations to converge	8
Model status	Successfully trained (no errors)

4. Predictions & Prediction Probabilities (Sample of Test Set)

Below is a sample of 15 test-set applications showing the actual outcome, the model's predicted class, and the predicted probability for each class (via `model.predict()` and `model.predict_proba()`).

Actual	Predicted	Prob_Rejected(0)	Prob_Approved(1)
0.0	0.0	0.949	0.051
1.0	1.0	0.1064	0.8936
1.0	1.0	0.274	0.726
1.0	1.0	0.2151	0.7849
1.0	1.0	0.1017	0.8983
0.0	1.0	0.2053	0.7947
1.0	1.0	0.1251	0.8749
1.0	1.0	0.1373	0.8627
1.0	1.0	0.3304	0.6696
1.0	1.0	0.2696	0.7304
1.0	1.0	0.121	0.879
1.0	1.0	0.1416	0.8584
1.0	1.0	0.1152	0.8848
1.0	1.0	0.2064	0.7936
0.0	0.0	0.9021	0.0979

5. Basic Training Results (Test Set Evaluation)

Metric	Value
Accuracy	85.37%
Precision	83.17%
Recall	98.82%
F1-score	90.32%
ROC-AUC	0.8536

Confusion Matrix:

	Predicted: Rejected	Predicted: Approved
Actual: Rejected	21	17
Actual: Approved	1	84

Key Observations:

- The model achieved 85.4% accuracy and 0.854 ROC-AUC on unseen test data, showing it generalizes well.
- Recall for Approved loans is very high (98.8%) — the model rarely misses a genuinely approvable applicant.
- Precision (83.2%) is a bit lower — a number of rejected applications were predicted as approved, mainly among applicants with good `Credit_History` but other weaker factors.
- Consistent with the earlier EDA, `Credit_History` remains the strongest driver behind the model's predictions.